AP - 18.3 - 21 14.13.2026

Changes in the level of RBC, can have significant effects on the body's ability to effectively deliver O_2 to the tissues. Ineffective hematopoises results in insufficient number of RBCs & results in one of several forms of anemia

{ polycythemia }

Percent saturation is normally monitered using a device known as a pulse oximeter which is applied to a thin part of the body → typically tip of finger. → sends red & infrared light through the finger → measuring the light with a photodetecter as it exits → Hemoglobin absorbs light differentially depending upon its saturation. with O_2!

The machine calibrates the amount of light received by the photodetector against the amount absorbed by the partially oxygenated hemoglobin & presents the data as percent saturation. Normal pulse oximeter readings range from 95-100 % →

lower percentages = hypoxemia / low blood O_2

generic term

~~At~~ Arterial stick

→ When this method is applied → the amount of O_2 present is expressed in terms of partial pressure of O_2 / PO_2

(P_{O_2}) → Recorded in units of mL in mercury (mm Hg)